

AI Technical Program Manager Handbook

Recommended Page Allocation

Chapter	Pages	Focus Area
1. AI Foundations	6	Core AI concepts and terminology
2. Generative AI	5	Enterprise GenAI adoption
3. Large Language Models	5	LLM architecture and concepts
4. Prompt Engineering	5	Prompt design techniques
5. Retrieval-Augmented Generation (RAG)	5	Enterprise knowledge systems
6. AI Agents	5	Agentic workflows and orchestration
7. AI Product Lifecycle	5	End-to-end AI delivery
8. Model Evaluation	4	Metrics and performance
9. System Design	6	Scalable AI architectures
10. Azure AI Architecture	6	Enterprise Azure patterns
11. Azure AI Foundry	4	AI development platform
12. Azure AI Search	4	Search and retrieval systems
13. Azure Functions	4	Serverless integrations
14. Azure DevOps	4	CI/CD and delivery management
15. AI Governance	5	Governance frameworks
16. Responsible AI	4	Ethics and compliance
17. Program Management for AI	6	TPM best practices
18. Stakeholder Management	4	Leadership and communication
19. Cost Estimation	4	Cloud and AI budgeting
20. AI Risk Management	4	Risk identification and mitigation
21. Migration Case Studies	6	Real-world enterprise examples
22. Executive Communication	3	Reporting and storytelling
23. Interview Preparation	5	TPM interview preparation
24. Leadership Stories	4	STAR leadership examples
25. AI TPM Career Roadmap	5	Career progression planning

Estimated Total Pages: 120

Suggested Chapter Breakdown

Chapter 1: AI Foundations

Sections

- Introduction to Artificial Intelligence
- AI vs Machine Learning vs Deep Learning
- Generative AI Overview
- Enterprise AI Adoption Trends
- AI TPM Responsibilities
- AI Maturity Models

TPM Deliverables

- AI Readiness Assessment
 - Stakeholder Mapping
 - Success Metrics Definition
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Chapter 2: Generative AI

Sections

- Evolution of Generative AI
- Text, Image, Audio, and Video Generation
- Enterprise Use Cases
- Security Considerations
- Governance Models

Enterprise Examples

- Customer Support Assistants
 - Code Generation Platforms
 - Knowledge Management Solutions
 - Document Automation
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Chapter 3: Large Language Models (LLMs)

Sections

- Transformer Architecture
- Tokens and Tokenization
- Embeddings
- Context Windows
- Fine-Tuning vs Prompt Engineering
- Model Selection Framework

TPM Perspective

Question	Example
Which model should we use?	GPT-5.5 vs Open Source
What is the cost impact?	Token consumption analysis
What are the risks?	Hallucinations and bias
How do we scale?	Multi-region deployment

Chapter 4: Prompt Engineering

Sections

- Zero-Shot Prompting
- One-Shot Prompting
- Few-Shot Prompting
- Chain of Thought
- Tree of Thought
- Structured Outputs
- Prompt Libraries

Deliverables

- Prompt Repository
- Prompt Review Checklist
- Evaluation Framework

Chapter 5: Retrieval-Augmented Generation (RAG)

Sections

- Why RAG Matters
- Enterprise Knowledge Systems
- Chunking Strategies
- Embedding Models
- Vector Databases
- Retrieval Methods
- Evaluation Metrics

Enterprise RAG Architecture

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graph TD
```

```
A[Enterprise Documents]  
--> B[Document Chunking]
```

```
B --> C[Generate Embeddings]

C --> D[Vector Database]

D --> E[Retriever Service]

E --> F[Azure OpenAI]

F --> G[Response Generation]

G --> H[End Users]
```

Azure Services

- Azure AI Search
- Azure OpenAI
- Azure Blob Storage
- Azure Functions
- Azure Monitor

Chapter 6: AI Agents

Sections

- What are AI Agents?
- Agentic AI Fundamentals
- Tool Calling
- Memory Management
- Multi-Agent Systems
- MCP (Model Context Protocol)
- Agent Governance

Enterprise Use Cases

- IT Helpdesk Agent
- Incident Response Agent
- Procurement Assistant
- TPM Dashboard Agent

Chapter 7: AI Product Lifecycle

Lifecycle Stages

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graph LR

A[Ideation]
--> B[Proof of Concept]
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B --> C[MVP]
C --> D[Pilot]
D --> E[Production]
E --> F[Optimization]

TPM Activities

Phase	TPM Responsibility
Ideation	Business Alignment
PoC	Scope Definition
MVP	Delivery Management
Pilot	Stakeholder Communication
Production	Governance
Optimization	KPI Tracking

Chapter 8: Model Evaluation

Evaluation Metrics

Metric	Description
Accuracy	Correct predictions
Precision	Relevant responses
Recall	Coverage of results
Latency	Response time
Cost	Token consumption
Reliability	Consistency of outputs

Chapter 9: System Design

Topics

- Scalability
- Availability
- Reliability
- Security
- Observability
- Disaster Recovery

Design Principles

1. Design for failure.
2. Optimize for scale.
3. Build for observability.
4. Secure by default.
5. Automate operations.

Chapter 10: Azure AI Architecture

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graph TD
  A[Users] --> B[Front-End Application]
  B --> C[API Management]
  C --> D[Azure OpenAI]
  D --> E[Azure AI Search]
  E --> F[Blob Storage]
  F --> G[Azure Monitor]
  G --> H[Operations Team]
```

Chapters 11–25 Overview

Chapter	Major Deliverables
Azure AI Foundry	Prompt Flows, Evaluations
Azure AI Search	Search Index Templates
Azure Functions	Serverless Playbooks
Azure DevOps	CI/CD Templates
AI Governance	Governance Charter
Responsible AI	Compliance Checklist
Program Management for AI	RAID Logs and OKRs
Stakeholder Management	Stakeholder Matrix
Cost Estimation	Budget Calculator
AI Risk Management	Risk Register

Chapter	Major Deliverables
Migration Case Studies	Lessons Learned Documents
Executive Communication	Steering Committee Templates
Interview Preparation	50 TPM Questions
Leadership Stories	STAR Framework Examples
AI TPM Career Roadmap	5-Year Growth Plan

Standard Chapter Components

Every chapter should include:

- Learning Objectives
 - Beginner Explanation
 - Intermediate Explanation
 - TPM Relevance
 - Enterprise Example
 - Azure Implementation
 - Mermaid Diagram
 - TPM Checklist
 - Templates
 - Common Mistakes
 - Key Takeaways
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Appendix

Templates Included

1. RAID Log
 2. Sprint Planning Template
 3. PI Planning Template
 4. Stakeholder Matrix
 5. RACI Matrix
 6. Executive Dashboard
 7. Cost Estimator
 8. AI Governance Checklist
 9. AI Readiness Assessment
 10. Interview Scorecard
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Footer Specification

Every page in the handbook must include:

This formatting ensures a consistent, corporate, PDF-ready handbook suitable for Program Managers, Technical Program Managers, Delivery Managers, Engineering Leaders, and aspiring AI TPMs.
